Momentum and Collisions

SS ILP Physics (Intermediate), Week 5 Class 2

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Outline

- Two-dimensional motion
- Internal versus External Forces
- Conservation of Momentum
- 2-D motion

Reference:
Physics with Applications by Douglas Giancoli (7th Edition)
Physics by J. S. Walker.

Summary (recap)

Momentum $\vec{P} = m\vec{v} \rightarrow P = mv$

Impulse $\vec{I} = \Delta\vec{P} = \vec{F}_{\text{avg}}\Delta t \rightarrow I = \Delta P = F_{\text{avg}}\Delta t$

Momentum and Force

$$\vec{F} = \frac{d\vec{P}}{dt} \rightarrow F = \frac{dP}{dt}$$

Two-dimensional motion

Momentum $\vec{P} = m\vec{v} \rightarrow P_x = mv_x$ and $P_y = mv_y$

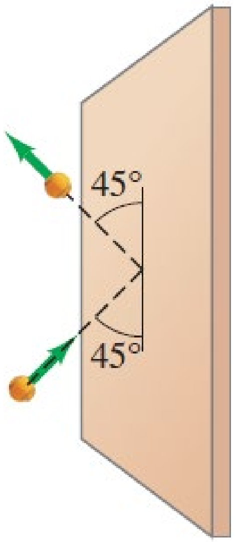
Impulse $\vec{I} = \Delta\vec{P} = \vec{F}_{\text{avg}}\Delta t \rightarrow I_x = \Delta P_x = F_{\text{avg},x}\Delta t$ and $I_y = \Delta P_y = F_{\text{avg},y}\Delta t$

Momentum and Force

$$\vec{F} = \frac{d\vec{P}}{dt} \rightarrow F_x = \frac{dP_x}{dt} \text{ and } F_y = \frac{dP_y}{dt}$$

Example:

A tennis ball of mass $m = 0.06\ kg$ and speed $v = 28$ m/s strikes a wall at a 45° angle and rebounds with the same speed at 45°, as shown in the figure below. What is the impulse (magnitude and direction) given to the ball?



Example: Solution

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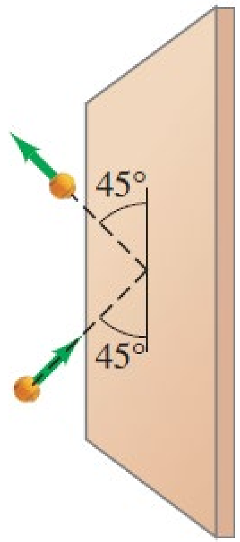


$$I_x = \Delta P_x = mv_{x,f} - mv_{x,i} = 0.06(-28\sin(45°)) - 0.06(28\sin(45°))$$
$$= -\frac{2(0.06)28\sqrt{2}}{2} = -2.4\ \text{kg m/s} = -2.4\ \text{N}\cdot\text{s}$$

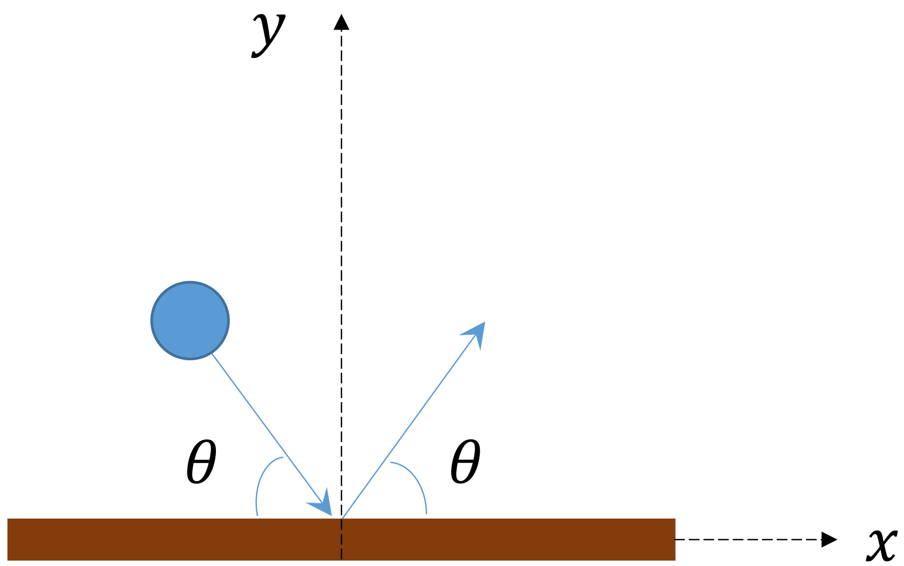
$$I_y = \Delta P_y = mv_{y,f} - mv_{y,i} = 0.06(28\cos(45°)) - 0.06(28\cos(45°)) = 0$$

$$\vec{I} = I_x\hat{\imath} + I_y\hat{\jmath} = (-2.4\hat{\imath})\ \text{N}\cdot\text{s}$$

As we specified that the positive direction is into (or right), the impulse is 2.4 kg m/s to the left.

Case Problem 1

A ball hits the ground and rebounds with the same speed, as shown below. What are the changes in the x and y components of the momentum of the ball?



Case Problem 1

Solution: As the ball hits and rebounds at the same angle, the momentum in the x-direction is the same.

For the momentum in the y-direction, we observe that the direction has changed, from going into the wall (negative) to going away from the wall (positive).

Thus, $\Delta p_x = 0$ and $\Delta p_y = mv_{y,f} - mv_{y,i} = 2mv\sin\theta > 0$.

Internal versus External Forces

- The net force acting on a system of objects is the sum of forces applied from outside the system (external forces) and forces acting between objects within the system (internal forces).

$$\vec{F}_{net} = \sum \vec{F}_{ext} + \sum \vec{F}_{int}$$

- Internal forces always occur in action-reaction pairs.

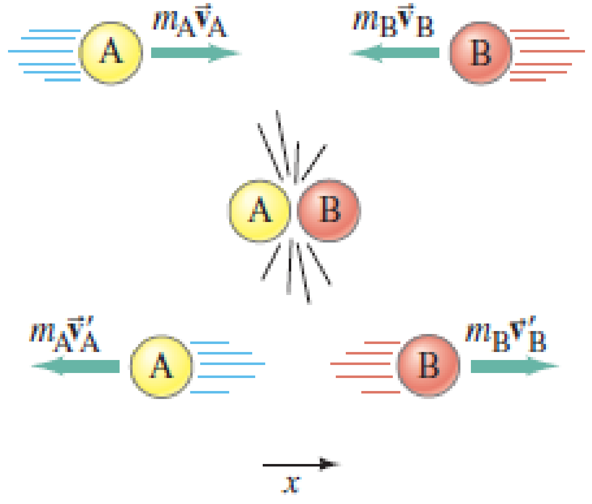
$$\sum \vec{F}_{int} = 0$$

$$\vec{F}_{net} = \sum \vec{F}_{ext}$$

J. S. Walker, Physics, p249

Conservation of Momentum

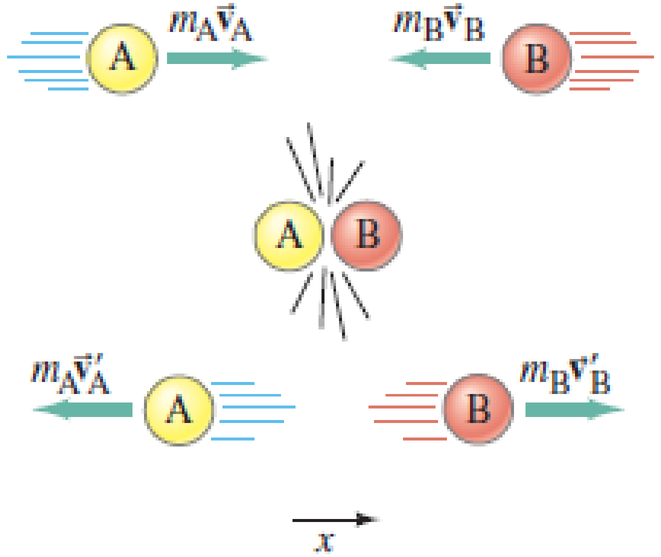
The concept of momentum is particularly important because, if no net external force acts on a system, the total momentum of the system is a conserved quantity.



Consider the head-on collision of two billiard balls, as shown on the right.

We assume the net external force on this system of two balls is zero.

Conservation of Momentum



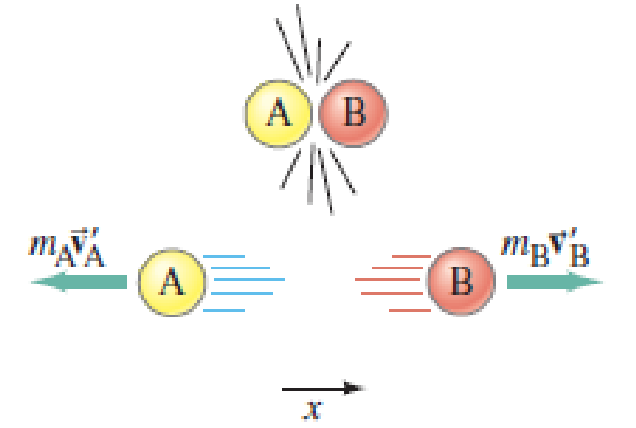
This means that the only significant forces during the collision are the forces that each ball exerts on the other.

Although the momentum of each of the two balls changes as a result of the collision, the sum of their momenta is found to be the same before as after the collision.

$$\Delta\vec{P} = 0 \;\rightarrow\; \vec{P}_f = \vec{P}_i \qquad \text{since } F = \frac{dP}{dt} = 0$$

Conservation of Momentum

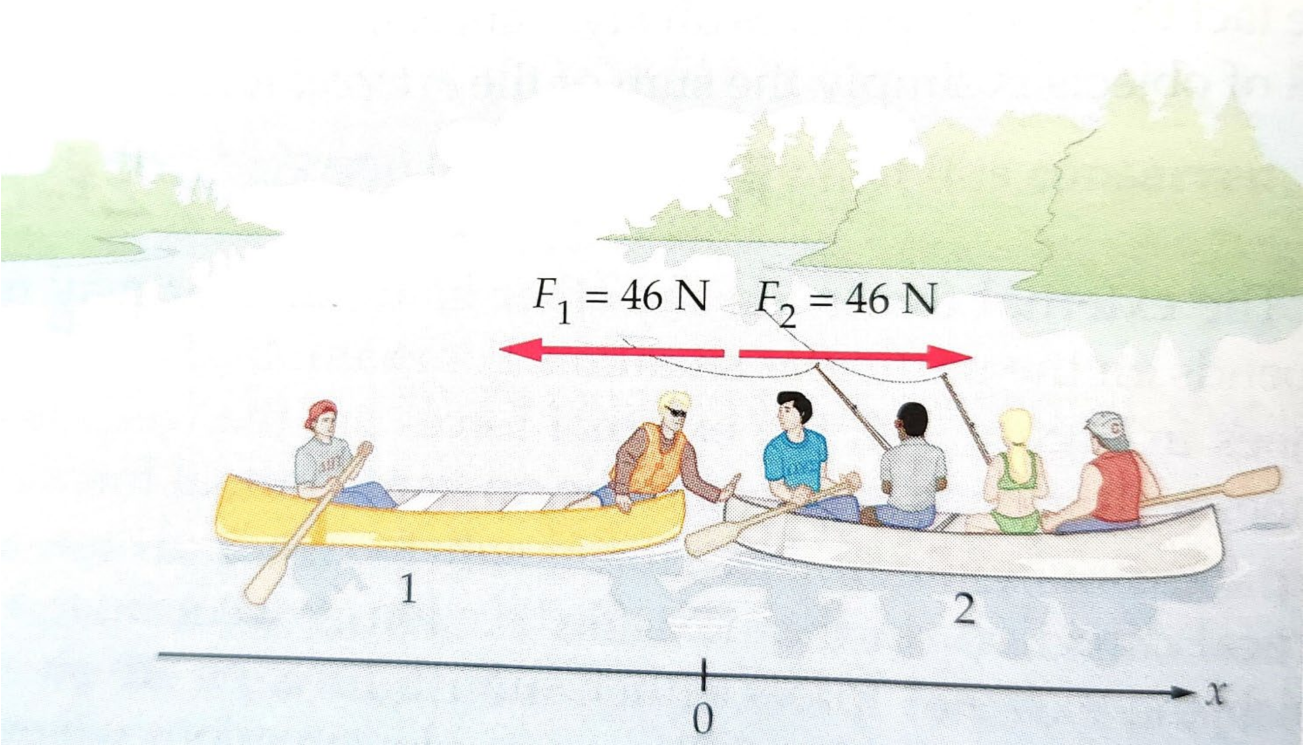
The key to solve conservation of momentum problems is remembering that momentum is a vector. Again, it is because momentum is derived from velocity and the velocity is a vector.



$\vec{P}_f = \vec{P}_i \;\rightarrow\; m_A\vec{v}_A + m_B\vec{v}_B = m_A\vec{v}'_A + m_B\vec{v}'_B$ where $\vec{v}_A$ and $\vec{v}_B$ represent the initial velocities of balls A and B respectively before the collision, $\vec{v}'_A$ and $\vec{v}'_B$ denote the final velocities of balls A and B after collision.

Concept Question 1:

Two groups of canoeists in the middle of a lake. After a brief visit, a person in canoe 1 pushes on canoe 2 with a force of 46 N separate the canoes. If we neglect the force due to water, is the total momentum of the system of two groups of canoeists conserved in x direction?



J. S. Walker, Physics, Example 9-3, p250

Concept Question 1: Solution

Two groups of canoeists in the middle of a lake. After a brief visit, a person in canoe 1 pushes on canoe 2 with a force of 46 N separate the canoes. If we neglect the force due to water, is the total momentum of the system of two groups of canoeists conserved in x direction?

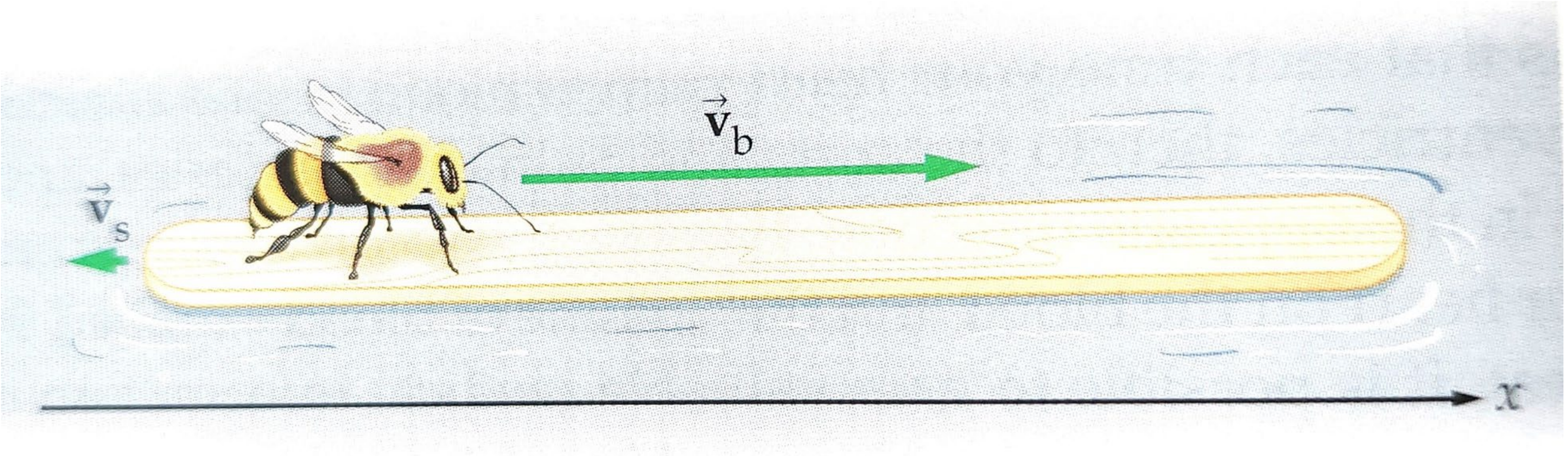
Yes, as there is no external force in x direction.

J. S. Walker, Physics, Example 9-3, p250

Restricted

Example

A honeybee with a mass of 0.15 g lands on one end of a floating 4.75-g popsicle stick. After sitting at rest for a moment, it runs toward the other end with a velocity $\vec{v}_b$ relative to the still water. The stick moves into the opposite direction with a speed of 0.12 cm/s. What is the velocity of the bee? Neglect the force due to water.



J. S. Walker, Physics, Active Example 9-2, p252

Example: Solution

A honeybee with a mass of 0.15 g lands on one end of a floating 4.75-g popsicle stick. After sitting at rest for a moment, it runs toward the other end with a velocity $\vec{v}_b$ relative to the still water. The stick moves into the opposite direction with a speed of 0.12 cm/s. What is the velocity of the bee? Neglect the force due to water.

$$m_b \vec{v}_{b,i} + m_s \vec{v}_{s,i} = m_b \vec{v}_{b,f} + m_s \vec{v}_{s,f}$$

$$m_b \times 0 + m_s \times 0 = m_b \vec{v}_{b,f} + m_s \vec{v}_{s,f}$$

$$0 = (0.15\ g) v_{b,f} + (4.75\ g)(-0.12\ cm/s)$$

$$v_{b,f} = \frac{(4.75\ g)(0.12\ cm/s)}{(0.15\ g)} = 3.8\ cm/s \quad \rightarrow \quad \vec{v}_{b,f} = (3.8\ cm/s)\ \hat{\imath}$$

J. S. Walker, Physics, Active Example 9-2, p252

Concept Question 2:

Two balls, of mass m and $2m$, collide and stick together. The combined balls are at rest after the collision. If the ball of mass m was moving 5 m/s to the right before the collision, what was the velocity of the ball of mass $2m$ before the collision?

(A) 2.5 m/s to the right
(B) 2.5 m/s to the left
(C) 10 m/s to the right
(D) 10 m/s to the left
(E) 1.7 m/s to the left

Concept Question 2: Solution

Answer is B.

The total momentum after collision after collision is zero because we are told that the combined balls are at rest. So the total momentum before collision must be zero as well. Let the direction to the right be positive. Let v be initial velocity of the ball of mass 2*m.*

$$m_{left}(5) + m_{right}(v) = m(5) + (2m)(v) = 0 \quad \rightarrow \quad v = -2.5\ \mathrm{m/s}$$

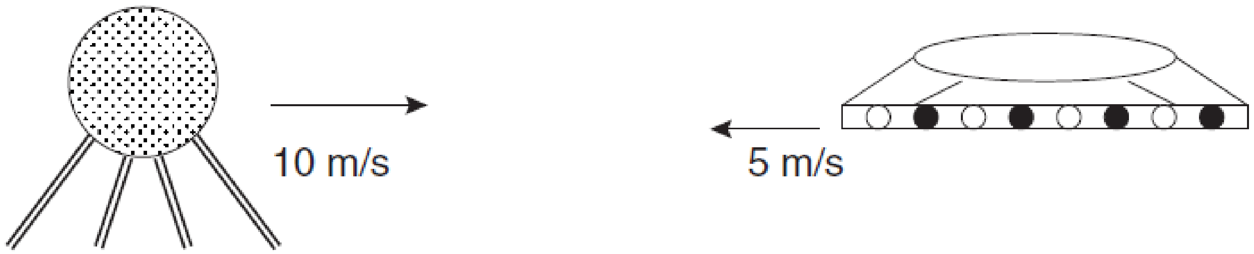
Note the minus sign which means that it is moving opposite to the direction to ball of mass *m*, i.e. to the left.

Case Problem 2:

A satellite floating through space collides with a small UFO. Before the collision, the satellite was traveling at 10 m/s to the right, and the UFO was traveling at 5 m/s to the left. Assume the satellite's mass is 70 kg and the UFO's mass is 50 kg.

(i) If the satellite and the UFO bounce off each other upon impact, what is the satellite's final velocity if the UFO has a final velocity of 3 m/s to the right?

(ii) If the satellite and UFO get stuck together upon colliding, then what is the final velocity of both objects?



Case Problem 2: Solution

Solution: First, we need to specify which direction is positive. We shall choose the initial direction of travel to the right.

(i) Next, we apply the conservation of momentum principle:

$$m_{sat}\vec{v}_{sat} + m_{ufo}\vec{v}_{ufo} = m_{sat}\vec{v}'_{sat} + m_{ufo}\vec{v}'_{ufo}$$

$$\rightarrow \ (70)(10) + (50)(-5) = (70)v'_{sat} + (50)(3) \ \rightarrow \ v'_{sat} = 4.3 \text{ m/s}$$

Case Problem 2: Solution

(ii) $m_{sat}\vec{v}_{sat} + m_{ufo}\vec{v}_{ufo} = m_{sat}\vec{v}'_{sat} + m_{ufo}\vec{v}'_{ufo}$

where this time, we have $v'_{sat} = v'_{ufo} = v'_{combi}$

$$\rightarrow \quad (70)(10) + (50)(-5) = (70 + 50)v'_{combi} \;\rightarrow\; v'_{combi} = 3.8\text{ m/s}$$

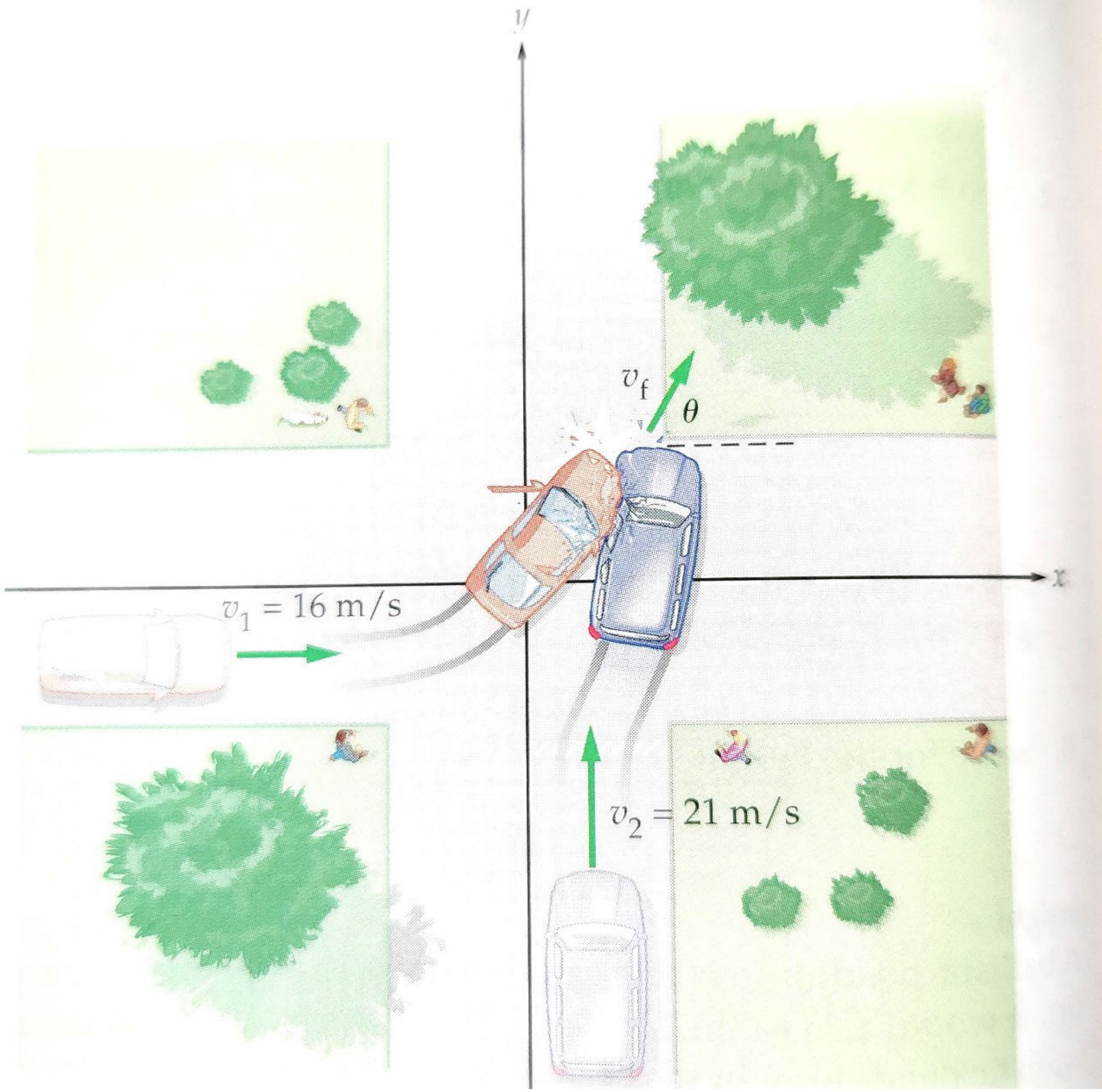
Two-dimensional Motion

In solving a two-dimensional motion problem, momentum is a vector and as a vector, it can be broken into x and y components. So, momentum conservation is conserved in each dimension.

$$\vec{P}_f = \vec{P}_i \Longrightarrow \begin{cases} \vec{P}_{f,x} = \vec{P}_{i,x} \\ \vec{P}_{f,y} = \vec{P}_{i,y} \end{cases}$$

Example

A car with a mass of 950 kg and a speed of 16 m/s approaches an intersection, as shown. A 1300-kg minivan travelling at 21 m/s is heading for the same intersection. The car and minivan collide and stick together. Find the speed and direction of the wrecked vehicles just after the collision, assuming external forces can be ignored.



J. S. Walker, Physics, Example 9-6, p257

Example: Solution

$$\vec{v}'_c = \vec{v}'_v = \vec{v}_f \qquad v'_{c,x} = v'_{v,x} = v_f \cos\theta$$

$$v'_{c,y} = v'_{v,y} = v_f \sin\theta$$

$$m_c\vec{v}_{c,x} + m_v\vec{v}_{v,x} = m_c\vec{v}'_{c,x} + m_v\vec{v}'_{v,x} \rightarrow \quad m_c v_{c,0} + m_v \times 0 = (m_c + m_v)v_f \cos\theta$$

$$m_c\vec{v}_{c,y} + m_v\vec{v}_{v,y} = m_c\vec{v}'_{c,y} + m_v\vec{v}'_{v,y} \rightarrow \quad m_c \times 0 + m_v v_{v,0} = (m_c + m_v)v_f \sin\theta$$

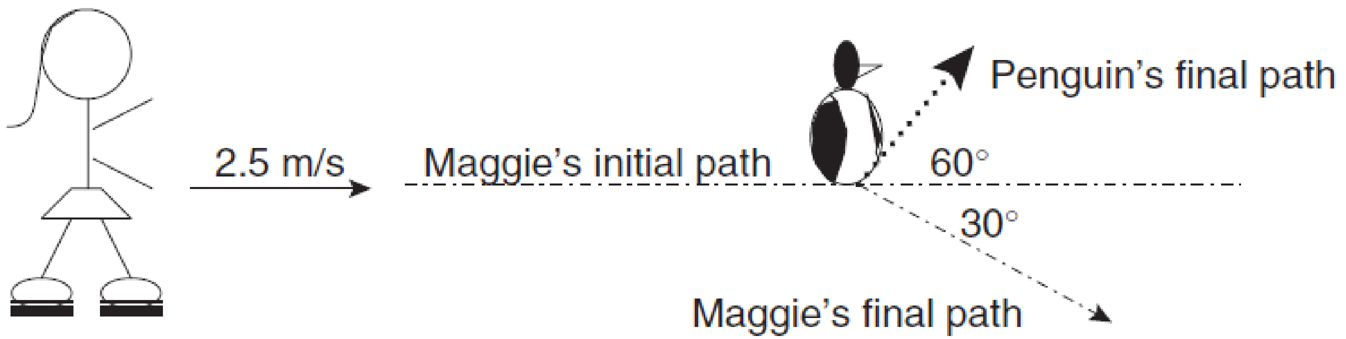
$$\tan\theta = \frac{m_v v_{v,0}}{m_c v_{c,0}} = \frac{(1300\ kg)(21\ m/s)}{(950\ kg)(16\ m/s)} = 1.8 \ \rightarrow \theta = \tan^{-1} 1.8 = 61^o$$

$$v_f = \frac{m_c v_{c,0}}{(m_c + m_v)\cos\theta} = \frac{(950\ kg)(16\ m/s)}{(950\ kg + 1300\ kg)\cos 61^o} = 14\ m/s$$

J. S. Walker, Physics, Example 9-6, p257

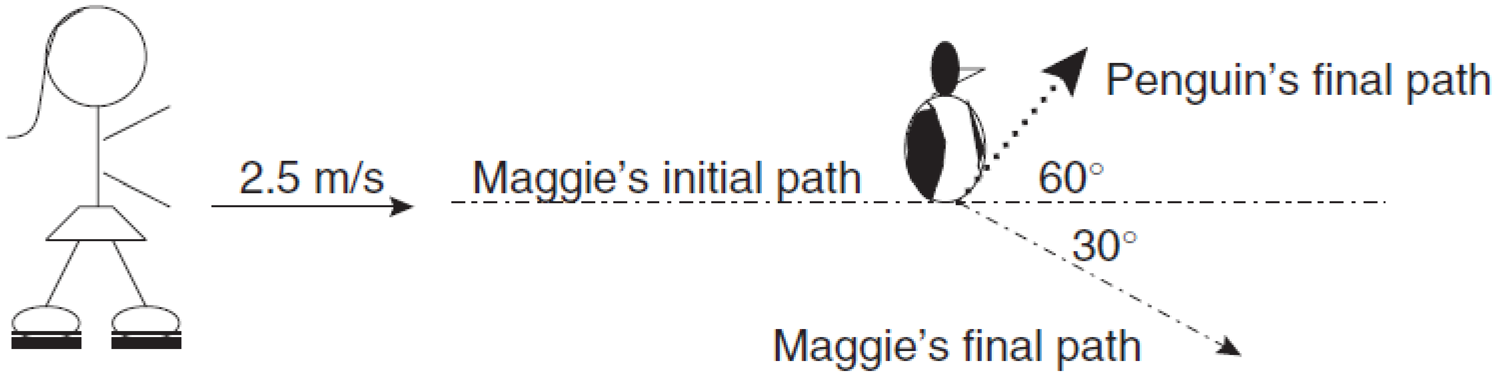
Case Problem 3:

Maggie has decided to go ice-skating. While cruising along, she trips on a crack in the ice and goes sliding. She slides along the ice at a velocity of 2.5 m/s. In her path is a penguin. Unable to avoid the flightless bird, she collides with it. The penguin is initially at rest and has a mass of 20 kg, and Maggie's mass is 50 kg. Upon hitting the penguin, Maggie is deflected 30° from her initial path, and the penguin is deflected 60° from Maggie's initial path. What is Maggie's velocity, and what is the penguin's velocity, after the collision?



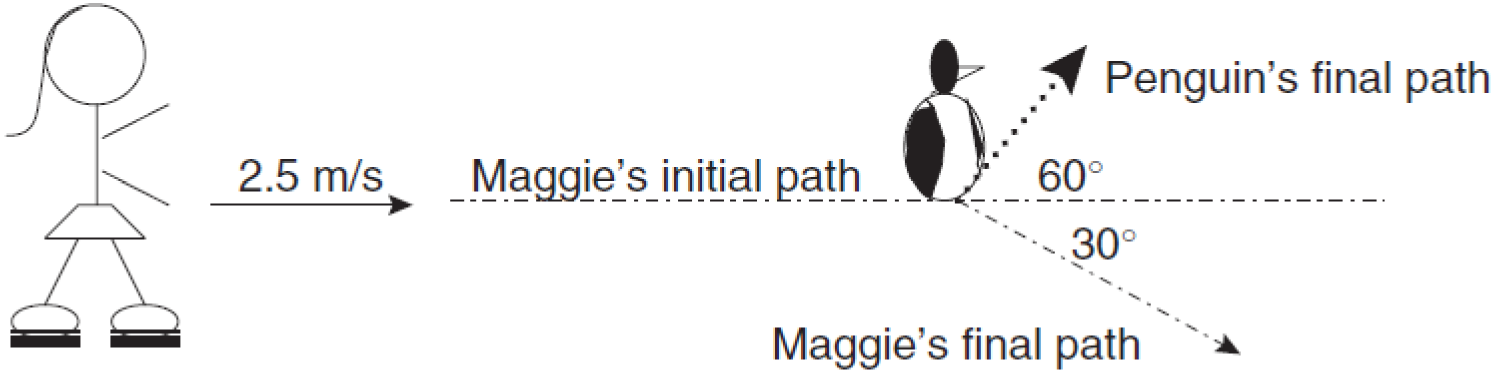
Case Problem 3: Solution

We need to analyze the momentum in the x and y directions separately and consider the momentum conservation individually!



We shall assume: (a) positive direction in x and y is right and upwards respectively. (b) After the collision, Maggie and the Penguin move as shown above. We can make this assumption provided we are consistent in its usage

Case Problem 3: Solution



throughout the solution. If we end up with a negative answer then we know the directions should have been reversed. The question wants us to find Maggie's and the Penguin's after the collision. So we are going to exploit trigonometry to find the x and y components of the velocity.

Case Problem 3: Solution

x-component equation for momentum conservation:

$$m_{Maggie}\vec{v}_{Maggie,x} + m_{Penguin}\vec{v}_{Pengin,x}$$
$$= m_{Maggie}\vec{v}'_{Maggie,x} + m_{Penguin}\vec{v}'_{Penguin,x}$$

$$50(2.5) + 20(0) = 50(v'_{Maggie} \cdot \cos(30°)) + 20(v'_{Penguin} \cdot \cos(60°))$$

$$\rightarrow \quad 125 = 50(v'_{Maggie} \cdot \sqrt{3}/2) + 20(v'_{Penguin} \cdot 1/2)$$

Case Problem 3: Solution

y-component equation for momentum conservation:

$$m_{Maggie}\vec{v}_{Maggie,y} + m_{Penguin}\vec{v}_{Pengin,y}$$
$$= m_{Maggie}\vec{v}'_{Maggie,y} + m_{Penguin}\vec{v}'_{Penguin,y}$$

$$50(0) + 20(0) = 50(-v'_{Maggie} \cdot \sin(30°)) + 20(v'_{Penguin} \cdot \sin(60°))$$

$$\rightarrow \qquad 0 = 50(-v'_{Maggie} \cdot 1/2) + 20(v'_{Penguin} \cdot \sqrt{3}/2)$$

Note that there is a negative sign for the y-component of Maggie's final velocity.

Case Problem 3: Solution

Solving both the x and y components of the momentum equation)

simultaneously,

$$125 = 50(v'_{Maggie} \cdot \sqrt{3}/2) + 20(v'_{Penguin} \cdot 1/2)$$

$$0 = 50(-v'_{Maggie} \cdot 1/2) + 20(v'_{Penguin} \cdot \sqrt{3}/2)$$

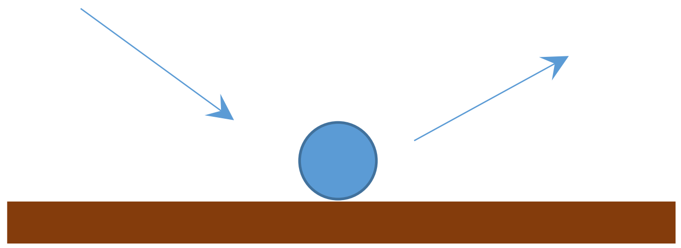
we would get $v'_{Maggie} = 2.17 \text{ m/s}$ and $v'_{Penguin} = 3.13 \text{ m/s}$

Off-class practice question(s)

Case Problem:

Given a ball of mass 0.25kg colliding the ground, it moves in at 12 m/s at $\theta_{in} = 30^o$ and moves out at 10 m/s at $\theta_{out} = 25^o$. The collision lasts for 10 ms, find

(a) the impulse delivered to the ball due to the collision, and
(b) the average force acting on the ball.



https://www.kpu.ca/sites/default/files/Faculty%20of%20Science%20%26%20Horticulture/Physics/PHYS%201100%20Collision%20%26%20Momentum%20Solutions.pdf

Case Problem: Solution

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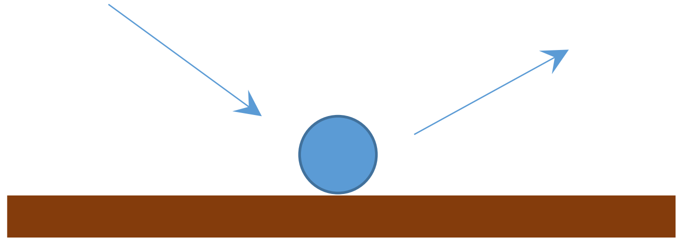
(a) the impulse delivered to the ball due to the collision, and
(b) the average force acting on the ball.

$$I_x = \Delta P_x = mv_{x,f} - mv_{x,i} = -0.33\ \mathrm{N \cdot s}$$

$$I_y = \Delta P_y = mv_{y,f} - mv_{y,i} = 2.56\ \mathrm{N \cdot s}$$

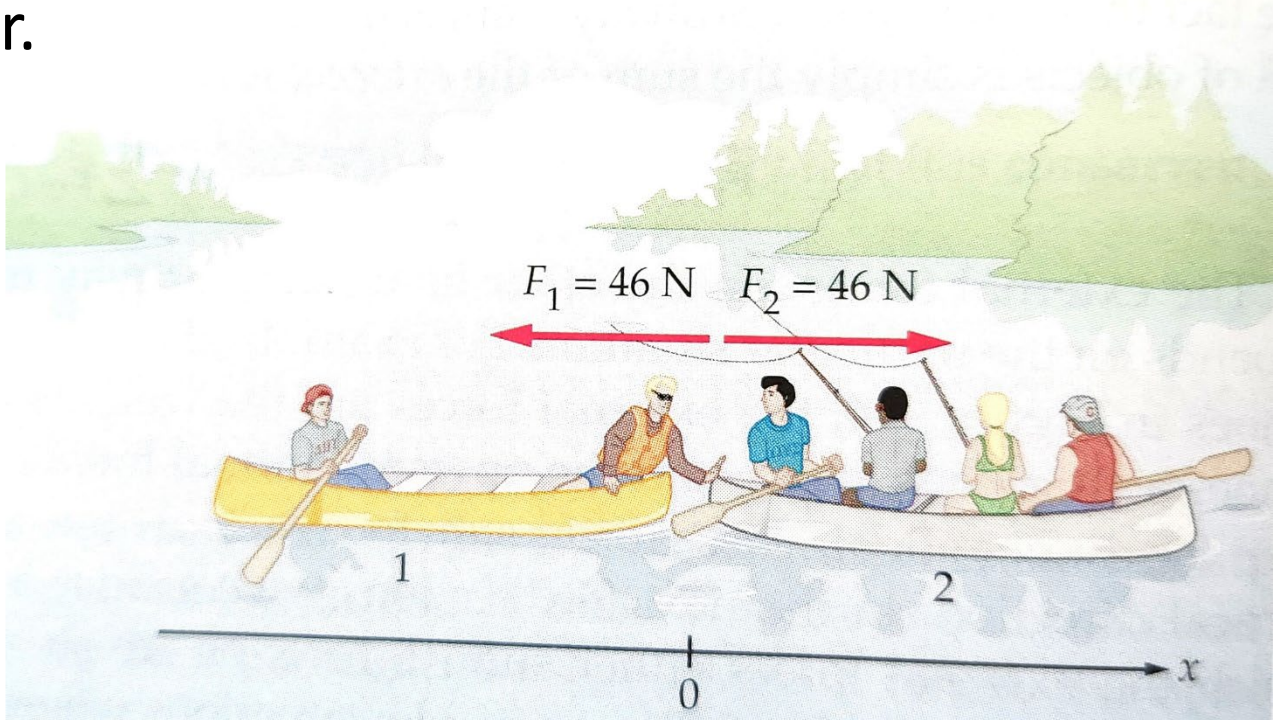
$$\vec{I} = I_x\hat{\imath} + I_y\hat{\jmath} = (-0.33\ \hat{\imath} + 2.56\hat{\jmath})\ \mathrm{N \cdot s}$$

$$\vec{F}_{\mathrm{avg}} = \frac{\vec{I}}{\Delta t} = \frac{(-0.33\ \hat{\imath} + 2.56\hat{\jmath})\ \mathrm{N \cdot s}}{10 \times 10^{-3} s} = (-33\ \hat{\imath} + 256\hat{\jmath})\mathrm{N}$$



Example

Two groups of canoeists in the middle of a lake. After a brief visit, a person in canoe 1 pushes on canoe 2 with a force of 46 N to separate the canoes. If the mass of canoe 1 and its occupants is 130 kg, and the mass of canoe 2 and its occupants is 250 kg, find the momentum of each canoe after 1.20 s of pushing. Neglect the force due to water.



J. S. Walker, Physics, Example 9-3, p250

Example: Solution

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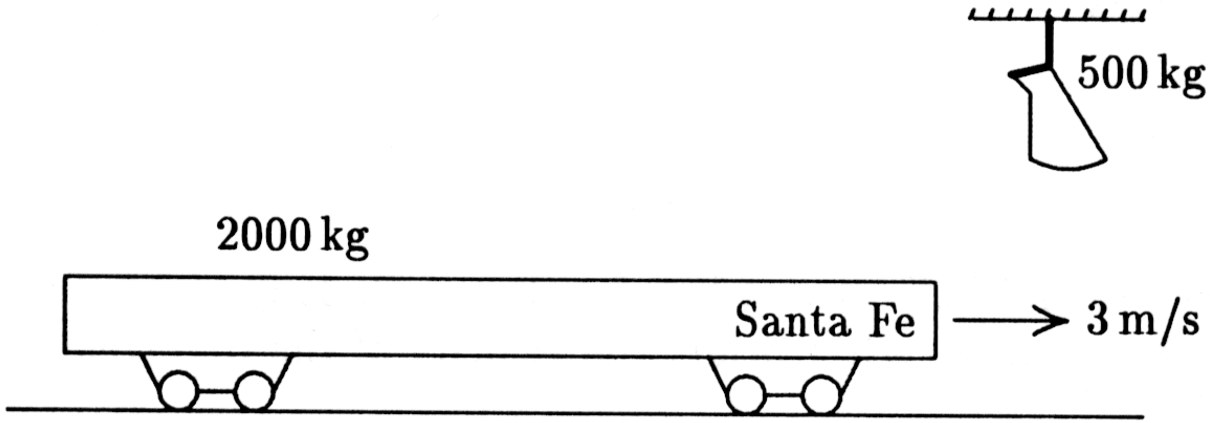
$$a_{1,x} = \frac{F_{1,x}}{m_1} = -\frac{0.35m}{s^2} \rightarrow \ v_{1,x} = a_{1,x}t = -0.42m/s \rightarrow P_{1,x} = m_1 v_{1,x} = -55\,kg \cdot m/s$$

$$a_{2,x} = \frac{F_{2,x}}{m_2} = 0.18\frac{m}{s^2} \rightarrow \ v_{2,x} = a_{2,x}t = 0.22\,m/s \rightarrow P_{2,x} = m_2 v_{2,x} = 55\,kg \cdot m/s$$

J. S. Walker, Physics, Example 9-3, p250

Case Problem

1. A 500-kg sack of coal is dropped on a 2000-kg railroad flatcar which was initially moving at 3 m/s as shown. After the sack rests on the flatcar, what is the speed of the flatcar?



Case Problem: Solution

Solution: We have to use the conservation of momentum principle.

$$m_{car}\vec{v}_{car,i} + m_{coal}\vec{v}_{coal,i} = m_{car}\vec{v}_{car,f} + m_{coal}\vec{v}_{coal,f}$$

$$2000(3) + 500(0) = 2000(v_{car,f}) + 500\left(v_{coal,f}\right)$$

Since the sack of coal eventually rests on the flatcar, we have $v_{car,f} = v_{coal,f}$

and this gives us

$$v_{car,f} = \frac{2000(3)}{2000+500} = \frac{12}{5} = 2.4 \text{ m/s}$$